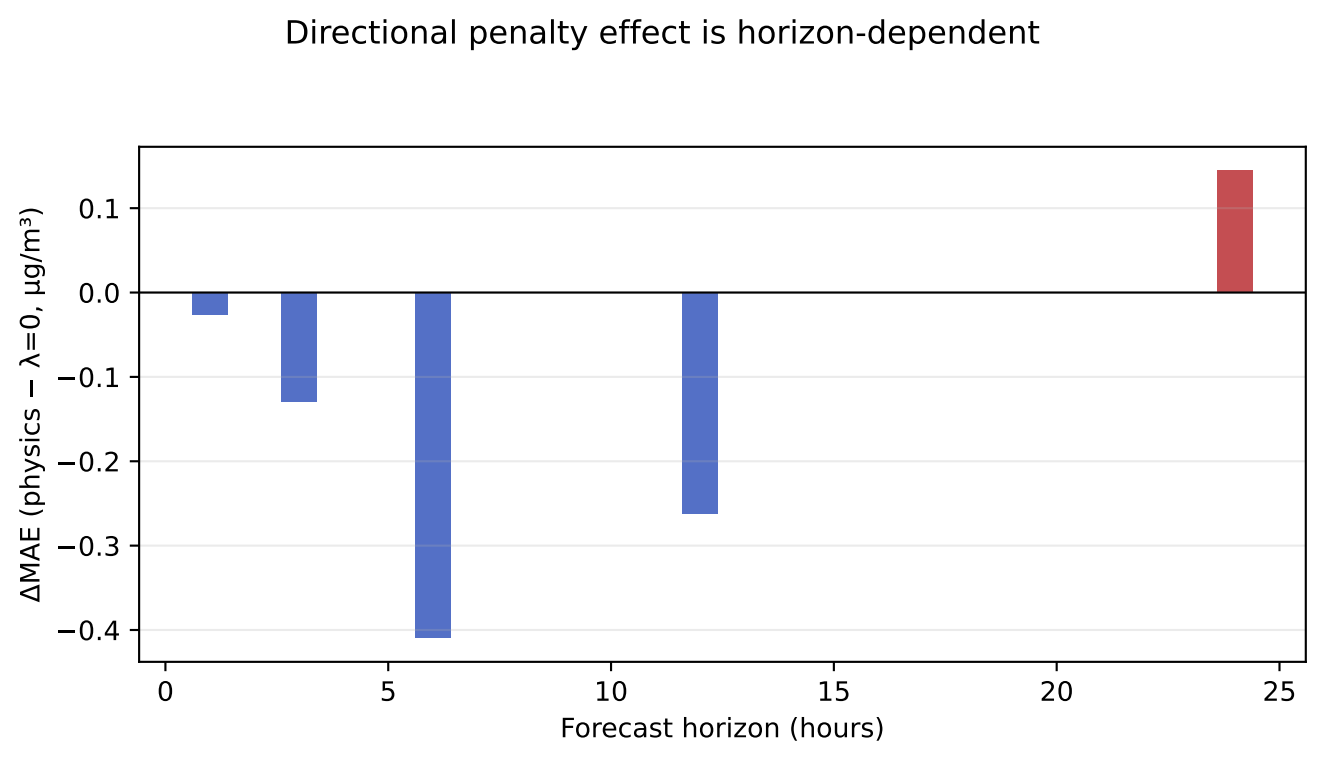
Directional penalty effect is horizon-dependent
ΔMAE (physics − λ=0, µg/m³)
0.1
0.0
−0.1
−0.2
−0.3
−0.4
0
5
10
15
20
25
Forecast horizon (hours)